A cooperative energy pool with optimised shared storage and Shapley settlement for the ComErgy benchmark community

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Abstract

The ComErgy Benchmark Dataset 2026 describes nine Sydney households with 49.3 kWp of rooftop solar and 65.2 kWh of batteries under a three-tier time-of-use tariff. This entry runs them as one cooperative pool. Solar serves all loads first, the fourteen battery modules act as one shared store scheduled by a daily linear programme, and the grid is used last. Economic performance: the pool pays 7,759 AUD per year against 12,425 AUD for the households alone, a saving of 4,666 AUD or 37.6 %. Community benefit and energy justice: the saving is divided by the Shapley value computed exactly over all 511 coalitions, with one tenth set aside for the two lower-income members; every household gains, by 16 % to 99 % of its own bill. Grid and environmental performance: 54 % of the solar is used locally and 60.5 % of demand is met locally. Import in the 0.55 AUD/kWh window falls from 916 to 63 kWh, and the pooled import never exceeds the community's own peak of 28.0 kW; halving that peak costs 116 AUD per year. Innovation: the exact settlement over every coalition, the zero-cost import cap and a workbook-driven model replace the approximations used at larger scale. Practicality: the pool needs only interval meters, hybrid inverters and one operator, and keeps 96 % of its saving without grid charging. Communication and reproducibility: every household receives a one-page statement, and every number is recomputed from the benchmark data with one command.

1 Introduction

Households with rooftop solar and batteries can serve each other before they serve the grid. Two questions must be settled for that to work: how the shared assets are run each hour, and how the money saved is divided so that every household is better off and the division can be explained. The ComErgy Benchmark Dataset 2026 poses these questions for nine Sydney households with different demand, solar and battery assets, a three-tier time-of-use tariff and a flat feed-in rate [1].

Published community energy studies answer one question at a time. Dispatch studies schedule shared batteries with optimisation or model predictive control and report the community cost [2]; allocation studies compare rules such as the Shapley value, the nucleolus and proportional sharing on a fixed set of coalition values [3, 4, 5]. Because the Shapley value needs the value of every coalition, larger studies approximate it by sampling or clustering [5], and the dispatch behind each coalition value is rarely the same as the one used for the community. The challenge also judges entries on six criteria at once: economic performance, fairness, grid and environmental performance, innovation, practicality and reproducibility [1]. The question this entry answers is therefore narrow: can a nine-household pool be dispatched and settled so that the community saving is close to its maximum, every household gains, the grid peak does not rise, and every number can be reproduced from the benchmark data?

The answer is a cooperative pool (Figure **1**). Each hour, solar generation serves all loads first. The fourteen battery modules act as one shared store, scheduled by a daily linear programme that plans two days ahead. The grid is used last, and the pooled import never exceeds the peak the community would have had without cooperation. Each household's benefit is measured against the bill it would pay alone with its own assets, so nobody can be worse off by definition. The annual saving is divided by the Shapley value, computed exactly over

all 511 coalitions of the nine households, with one tenth set aside for lower-income members. Every household receives a one-page statement.

The design determines the five elements the challenge lists [1]. Generation, storage and exchange within the community are handled by the pool and its shared store (Section 2.4). Purchase from and export to the grid follow the optimised dispatch under the tariff, the feed-in rate and the import cap (Section 2.4). Local trading and local prices are settled once a year, with an illustrative hourly price (Sections 2.5 and 2.7). Solar and batteries are coordinated by the daily programme (Section 2.4). Benefits are allocated by the Shapley-based rule (Section 2.5). Table C**1** in Appendix C maps every question of the challenge brief to the section that answers it.

The contribution is the combination at a scale where the exact cooperative solution is affordable. With nine households every coalition is dispatched with the same optimiser as the community, so the settlement rests on true marginal contributions, and the whole model runs from the benchmark data and one inputs workbook. Section 2 gives the method, Section 3 the results and their discussion, and Section 4 the conclusions against the six criteria. Appendix A lists the assumptions, Appendix B the nine household statements, Appendix C the mapping to the challenge brief and Appendix D the reproduction steps.

2 Methodology

2.1 Benchmark community, data and assumptions

The dataset provides 8,760 hourly values of demand for each of nine households, the import price, the feed-in rate, dry-bulb temperature and global horizontal irradiance, together with each household's solar and battery modules, type, occupancy and after-tax income [1]. Table **1** summarises the members. The year is 2021 by inference: the 0.55 AUD/kWh block is absent on 2 and 3 January and on every seventh day thereafter, which is a weekend only if 1 January is a Friday. The import price is 0.15 AUD/kWh overnight, 0.25 in the shoulder hours and 0.55 on weekdays from 13:00 to 19:00. The feed-in rate is 0.05 AUD/kWh throughout, and each household pays a connection fee of 1 AUD per day. Annual demand is 72,566 kWh with a coincident peak of 28.0 kW, against 55.9 kW for the sum of the individual peaks. Solar modules are rated 440 W and battery modules 4.66 kWh [1]; the community holds 112 solar and 14 battery modules.

Table 1. Members of the benchmark community [1].

Member	Household	Occupants	After-tax income (AUD)	Solar (kWp)	Battery (kWh)	Demand (kWh/y)	Peak (kW)
1	Couple, both full-time	2	200,000	6.6	0	5,316	4.3
2	Retired couple	2	80,000	0	9.3	4,401	4.0
3	Couple, three children	5	160,000	13.2	14.0	9,260	7.4
4	Couple, two children	4	240,000	9.7	0	7,131	6.9
5	Single parent, two children	3	70,000	0	0	6,921	3.5
6	Couple, three children	5	120,000	8.8	9.3	7,436	10.0
7	Multi-generation home	7	285,000	0	18.6	10,748	5.7
8	Couple, four children	6	145,000	6.6	14.0	10,463	6.9
9	Share house, seven adults	7	190,000	4.4	0	10,891	7.1
All		**41**		**49.3**	**65.2**	**72,566**	**28.0 (coincident)**

No data beyond the benchmark are used. Everything the dataset leaves open is an assumption with a cited basis: the solar orientation and losses, the battery power, efficiency, usable window and wear cost, the operator fee, the solidarity rule and the grid emission factor. Table **2** lists them and Appendix A (Table A**1**) gives the reasoning. Battery values follow the datasheet of the module named in the benchmark [6]. The wear cost is set at the lower end of the surveyed range, 0.065 AUD per kWh discharged; the upper end, 0.18 AUD per kWh, cuts the community saving by 22 % (Section 3.5).

Table 2. Parameters beyond the benchmark, with sources (full register in Appendix A and in the inputs workbook).

Group	Parameter	Value	Source
Solar	tilt, azimuth	30 degrees, north	[7, 8, 9]
Solar	albedo	0.20	[10]
Solar	temperature coefficient, NOCT	-0.33 %/K, 44 C	[11]
Solar	DC losses, inverter efficiency, DC to AC ratio	0.86, 0.97, 1.33	[12, 13, 14]
Battery	power per kWh of capacity	0.48 kW	[6, 15]
Battery	round-trip efficiency, usable window	0.95, 10 to 100 %	[6, 16, 17]
Battery	cycling cost	0.065 AUD/kWh discharged	[16, 18, 19, 20]
Tariff	import price, feed-in rate, connection fee	0.15/0.25/0.55 and 0.05 AUD/kWh, 1 AUD/day	[1, 21]
Community	operator fee	50 AUD per household per year	[22, 23, 24, 25, 26]
Settlement	solidarity share, income test	10 %, 108,000 AUD	[3, 27]
Grid	emission factor	0.79 kg CO2-e/kWh	[28, 29]
Dispatch	window, commitment, solver	48 h, 24 h, dual simplex	[2, 30]

The pipeline of Figure **1**(b) runs in five stages. Solar generation is derived from the benchmark weather (Section 2.2). The stand-alone bill of each household is computed (Section 2.3). The pool and every coalition are dispatched with the same optimiser (Section 2.4). The saving is settled (Section 2.5). Indicators, scenarios and the sensitivity are evaluated (Section 2.6). Symbols are defined in the Nomenclature. Units are kW, kWh and AUD. The time step is one hour, h = 1 to 8,760; households are indexed i = 1 to 9 and coalitions S.

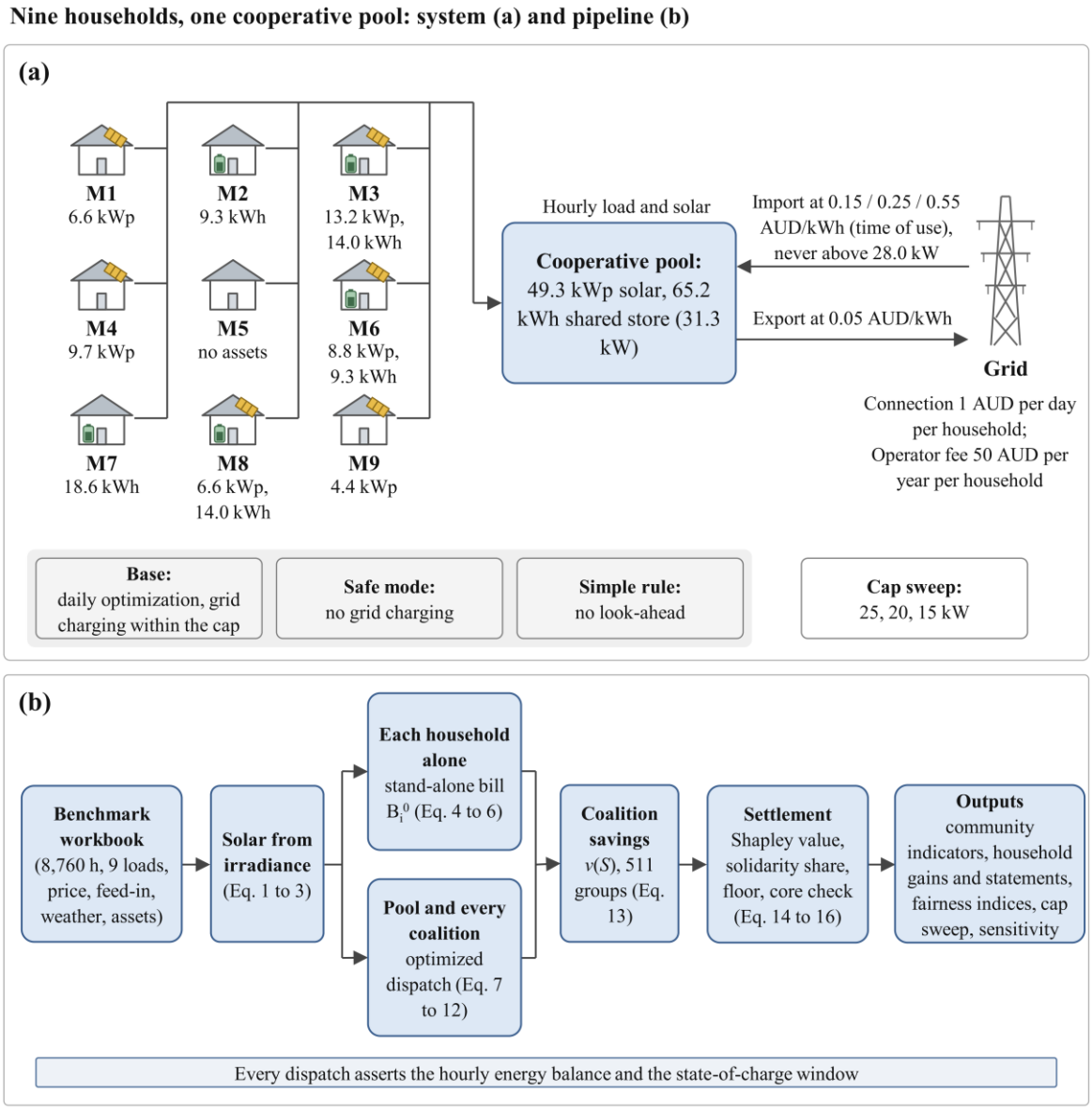


Figure 1. (a) The benchmark community of nine households with their rooftop solar and battery modules, the cooperative pool with its shared store, and the grid interface under the benchmark time-of-use tariff and feed-in rate; the strip lists the scenarios compared. (b) The computational pipeline from the benchmark workbook to the settlement outputs; equation numbers refer to Section 2.

2.2 Solar generation from the benchmark weather

The dataset gives horizontal irradiance and air temperature but no generation series, so generation per installed kilowatt is derived in three steps. The irradiance on the tilted panel follows from the Erbs split of the horizontal irradiance into beam and diffuse parts and an isotropic-sky transposition to a plane tilted 30 degrees towards north with albedo 0.20 [9, 10, 31, 32], Eq. (1):

$$POA(h) = DNI(h)\cos\theta(h) + DHI(h)\frac{1+\cos\beta}{2} + GHI(h)\rho\frac{1-\cos\beta}{2} \quad (1)$$

The cell temperature follows the nominal operating cell temperature relation. The output per kilowatt includes the temperature coefficient, the DC loss factor and the inverter efficiency [11, 12, 13, 14], Eq. (2):

$$y(h) = \frac{POA(h)}{1000}[1 + \gamma(T_c(h) - 25)]\eta_{dc}\eta_{inv}, T_c(h) = T(h) + \frac{NOCT - 20}{800}POA(h) \quad (2)$$

Household generation is the installed capacity times the yield, clipped at the inverter rating set by the DC to AC ratio [13], Eq. (3):

$$G_i(h) = min(k_i y(h), k_i/r) \quad (3)$$

The annual yield is 1,744 kWh per kWp before clipping and 1,694 kWh per kWp after it, 83,463 kWh in total. It is about 10 % above the long-term mean for Sydney because the benchmark irradiance (1,892 kWh/m2) is itself above that mean [7].

2.3 The stand-alone bill: what each household pays alone

Each household alone follows the self-consumption rule of a standard hybrid inverter, hour by hour and without look-ahead (Figure **2**, panel a). Solar serves the household's own load. Any surplus charges its own battery within the power limit, the one-way efficiency and the usable window. Any deficit is drawn from the battery. The grid supplies the remainder and buys the remaining surplus at the feed-in rate. The battery is never charged from the grid. For a surplus hour and a deficit hour the charge and discharge are given by Eq. (4), and the store follows Eq. (5):

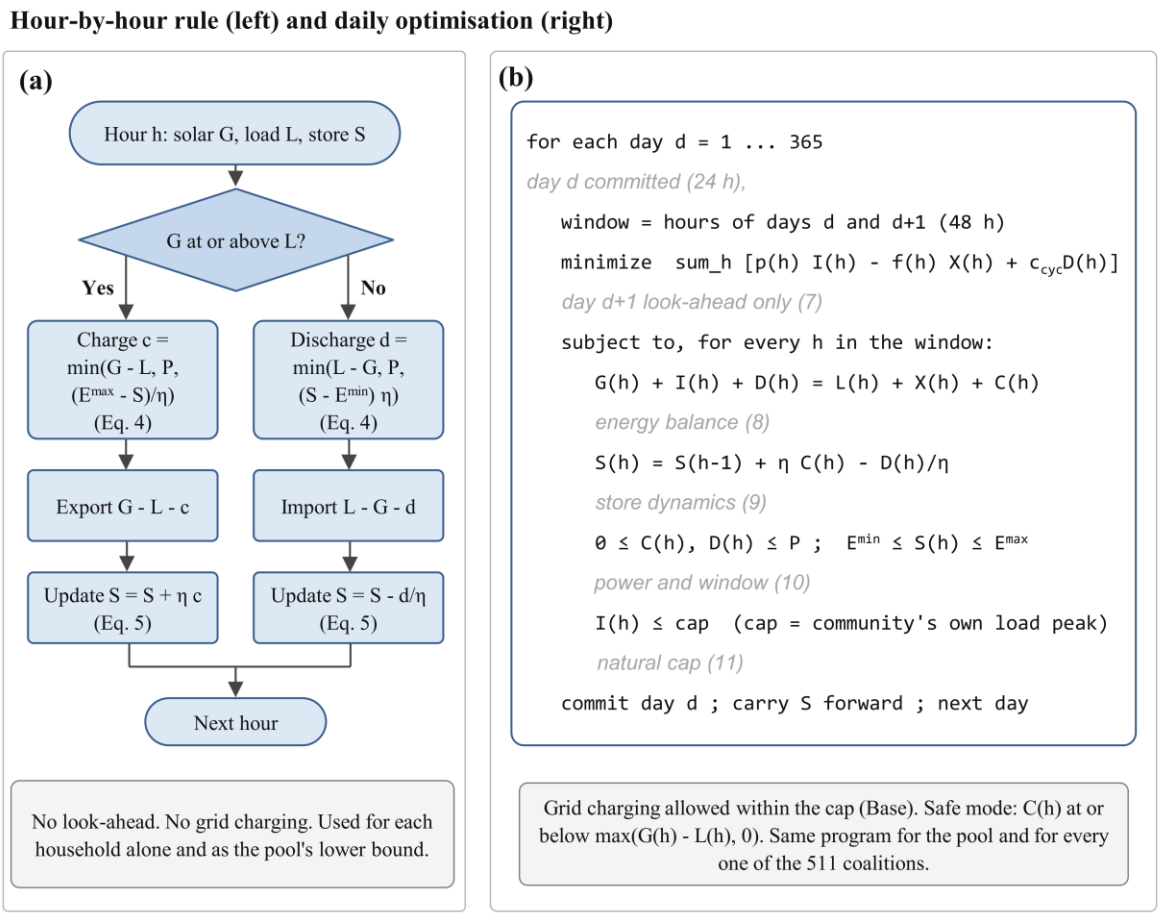
$$\begin{aligned} C_i(h) &= min(G_i - L_i, P_i, \frac{E_i^{max} - S_i(h-1)}{\eta}), D_i(h) \\ &= min(L_i - G_i, P_i, (S_i(h-1) - E_i^{min})\eta) \end{aligned} \quad (4)$$

$$S_i(h) = S_i(h-1) + \eta C_i(h) - D_i(h)/\eta \quad (5)$$

The stand-alone bill adds import cost, export revenue, a cycling cost on discharged energy and the connection fee, Eq. (6):

$$B_i^0 = \sum\nolimits_h [p(h)I_i(h) - f(h)X_i(h)] + c_{cyc}\sum\nolimits_h D_i(h) + 365c_{conn} \quad (6)$$

The stand-alone bill is the ceiling guaranteed to every household: under cooperation nobody pays more than it. Cooperative game theory calls this guarantee individual rationality, and the stand-alone outcome the disagreement point, the position each member falls back to if no agreement is reached [3, 4, 5]. The bill without any assets, $\sum_h p(h)L_i(h)$ plus the connection fee, is reported for context only.



Symbols: G solar, L load, I import, X export, C charge, D discharge, S state of charge, P power limit, E^{min} and E^{max} state-of-charge bounds, η one-way efficiency, p price, f feed-in, c_{cyc} cycling cost

Values (Table 2): η = 0.975 (round trip 0.95), P = 0.48 kW per kWh of store, E^{min} = 0.1 E and E^{max} = E (10 to 100 % window), c_{cyc} = 0.065 AUD/kWh; in the cap sweep the natural cap is replaced by 25, 20 and 15 kW.

Figure 2. The two dispatchers. (a) The self-consumption rule applied each hour without look-ahead, used for the stand-alone bills and as the lower bound for the pool. (b) The daily linear programme of the pool and of every coalition, solved over a two-day window with the first day committed; grid charging is allowed within the natural cap. Equation numbers refer to Section 2.

2.4 Pooled dispatch of the community and of every coalition

A coalition S of households is dispatched as one customer. Its load, generation and shared store are the sums over its members. Because all battery modules are identical, the single store reproduces the cost of the individual batteries exactly. Each day a linear programme over the next two days is solved and its first day is committed, with the state of charge carried forward. The second day is included only so that energy kept overnight has a use (Figure **2**, panel b) [2, 30]. With non-negative hourly variables for import, export, charge, discharge and state of charge, the programme minimises the cost of the window, Eq. (7):

$$min \sum_{h \in W} [p(h)I(h) - f(h)X(h) + c_{cyc}D(h)] \tag{7}$$

subject to the energy balance Eq. (8), the store dynamics Eq. (9), the power and window limits Eq. (10) and the import cap Eq. (11):

$$G_S(h) + I(h) + D(h) = L_S(h) + X(h) + C(h) \tag{8}$$

$$S(h) = S(h-1) + \eta C(h) - D(h)/\eta \tag{9}$$

$$0 \le C(h), D(h) \le P_S, E_S^{min} \le S(h) \le E_S^{max} \tag{10}$$

$$I(h) \le \bar{I}_S = max_h L_S(h) \tag{11}$$

Eq. (11) is the natural cap: the pooled import never exceeds the peak the coalition would have had without cooperation, so cooperation cannot worsen the network peak [33]. Grid charging is allowed within that cap. In the safe-mode scenario it is excluded by a further bound: the charge never exceeds the hour's own surplus. The programme has 240 variables per window and is solved 365 times per coalition with the dual simplex method. Every hour of every dispatch is checked against Eq. (8) to Eq. (10) by assertion. The self-consumption rule of Section 2.3 applied to the pool gives the lower bound of a simple controller.

The cost of a coalition adds the connection and operator fees of its members. The operator fee is the fixed running cost of the cooperative [22, 23, 24, 25, 26], Eq. (12):

$$c(S) = \sum_h [p(h)I_S(h) - f(h)X_S(h) + c_{cyc}D_S(h)] + |S|(365c_{conn} + c_{op}) \tag{12}$$

2.5 Settlement: how the saving is shared

The settlement treats the community as a cooperative game. A cooperative game assigns to every possible group of players, called a coalition, the value that group could secure on its own. Here the value of a coalition is its saving: what its members would pay alone minus what they pay together, Eq. (13):

$$v(S) = \sum_{i \in S} B_i^0 - c(S) \tag{13}$$

Eq. (13) is evaluated for all 511 non-empty coalitions with the same dispatch, which takes about eleven minutes on a desktop computer. The Shapley value is the standard rule for dividing the value of a cooperative game [34]. It gives each player the average amount by which the value grew when that player joined, taken over every order in which the group could have formed. It is used here for three reasons. It is the only rule that is efficient (the shares sum to the saving), symmetric (identical households get identical shares) and gives nothing to a household that adds nothing. It can be explained in one sentence. Small communities find it the easiest rule to accept [3, 4]. For household i it is Eq. (14):

$$\phi_i = \sum_{S \not\ni i} \frac{|S|!\,(8 - |S|)!}{9!} [v(S \cup \{i\}) - v(S)] \tag{14}$$

The recommended rule sets aside a solidarity share σ of the community saving $v(N)$, split equally among the n_{low} members whose after-tax income is below the Greater Sydney median gross household income. The remainder is allocated by the Shapley value. A floor raises any negative gain to zero and rescales the rest so that the gains sum to $v(N)$ [3, 4, 27], Eq. (15):

$$g_i = (1 - \sigma)\phi_i + \frac{\sigma v(N)}{n_{low}} \mathbb{1}\{y_i < y^*\}, g_i \leftarrow max(g_i, 0) \tag{15}$$

The settled bill is the stand-alone bill minus the gain. Settlement is annual, with one statement per household [4]. The same-percentage rule, which gives every household the same share of its own bill, is the baseline for comparison. Two stability checks are reported. An allocation is in the core when no coalition could do better by leaving the community; the maximum core excess, Eq. (16), is the largest amount any coalition could gain by leaving, and a value at or below zero means the allocation is in the core [5]. The nucleolus is the allocation that makes the most dissatisfied coalition as satisfied as possible. It is always in the core when the core is non-empty [35]:

$$e^{max}(g) = max_{S \subset N} [v(S) - \sum_{i \in S} g_i] \tag{16}$$

2.6 Indicators, scenarios and sensitivity

Fairness is reported as each household's relative saving (gain divided by stand-alone bill), its minimum, the minimum absolute gain, the Gini coefficient of the relative savings and the individual-rationality flag (all gains at or above zero). The Gini coefficient measures how unequal a set of shares is: 0 when all households save the same share of their bill and 1 when one household takes everything. The Lorenz curve plots the cumulative share of savings against the cumulative share of households, so an equal division lies on the diagonal and the Gini coefficient is twice the area between the curve and the diagonal. Community indicators are import and export, self-consumption (one minus export over generation), self-sufficiency (one minus import over load) and peak import. They also include equivalent full cycles (discharged energy over usable capacity), import inside the 0.55 window, and emissions of imports at the NSW grid factor of 0.79 kg CO2-e per kWh for 2021 [28, 29]. Table **3** lists the scenarios. Within the Base scenario the natural cap is replaced by explicit caps of 25, 20 and 15 kW. A one-at-a-time sensitivity moves the import price, the feed-in rate, the cycling cost, the operator fee and the solar yield across the ranges of Table **2** and records the community saving, the Gini coefficient and the minimum relative saving. It uses the self-consumption rule for all coalitions so that each evaluation takes seconds.

Table 3. Scenarios reported.

Scenario	Pool dispatch	Grid charging	Coalition dispatch	Purpose
Base	daily linear programme, two-day window	allowed within the natural cap	same programme	primary result
Safe mode	daily linear programme	not allowed	same programme	operation without grid charging
Simple rule	self-consumption rule, no look-ahead	not allowed	same rule	lower bound of a plug-and-play controller
Cap sweep	Base with explicit caps 25, 20, 15 kW	allowed within the cap	not needed	price of peak reduction

2.7 Local prices and household statements

Inside the pool there is no hour-by-hour trade between households. Shared energy is settled once a year through the rule of Section 2.5, so no household needs to watch a price. An hourly internal price is still computed for information, from the supply-demand ratio of the hour [36]. It equals the feed-in rate when local solar exceeds local load, rises towards the import price as the surplus shrinks, and equals the import price when there is no surplus. It is written beside the hourly dispatch and does not affect the settlement. Each household receives a one-page statement (Appendix B). The statement shows the bill without assets, the stand-alone bill, the pooled share, the saving, what the household contributed, its Shapley share in words, the solidarity amount where it applies, and the community totals.

3 Results and Discussion

3.1 Community level

Table **4** gives the community indicators of the three scenarios, and Figure **3** shows the pooled operation in a summer and a winter week. The nine households alone pay 12,425 AUD per year with their own assets, and 22,613 AUD without any assets. The pool pays 7,759 AUD including nine connection fees and nine operator fees. The saving is 4,666 AUD per year, 37.6 % of the stand-alone total. The daily optimisation is worth 428 AUD per year over the simple rule. Grid charging within the natural cap is worth 196 AUD per year over the safe mode. Solar generation (83.5 MWh) exceeds demand (72.6 MWh), so 38.4 MWh are exported whatever the dispatch; self-consumption is 54 % and self-sufficiency 60.5 %. The optimised pool imports only 63 kWh inside the 0.55 window, against 916 kWh under the simple rule, and that is where most of its advantage lies. Emissions of imports are 22.6 t CO2-e per year and differ by less than 1 % across the scenarios, because imports differ little. The lever on emissions is the exported surplus, not the dispatch.

Table 4. Community indicators by scenario (annual; costs include connection and operator fees).

Indicator	Base	Safe mode	Simple rule
Community cost (AUD)	7,759	7,955	8,187
Saving against stand-alone bills (AUD, %)	4,666 (37.6 %)	4,470 (36.0 %)	4,238 (34.1 %)
Import (kWh)	28,665	28,495	28,495
Export (kWh)	38,435	38,435	38,435
Self-consumption	0.539	0.539	0.539
Self-sufficiency	0.605	0.607	0.607
Peak import (kW)	28.0	27.4	27.4
Equivalent full cycles	365	310	310
Import in the 0.55 window (kWh)	63	428	916
Emissions of imports (t CO2-e)	22.6	22.5	22.5

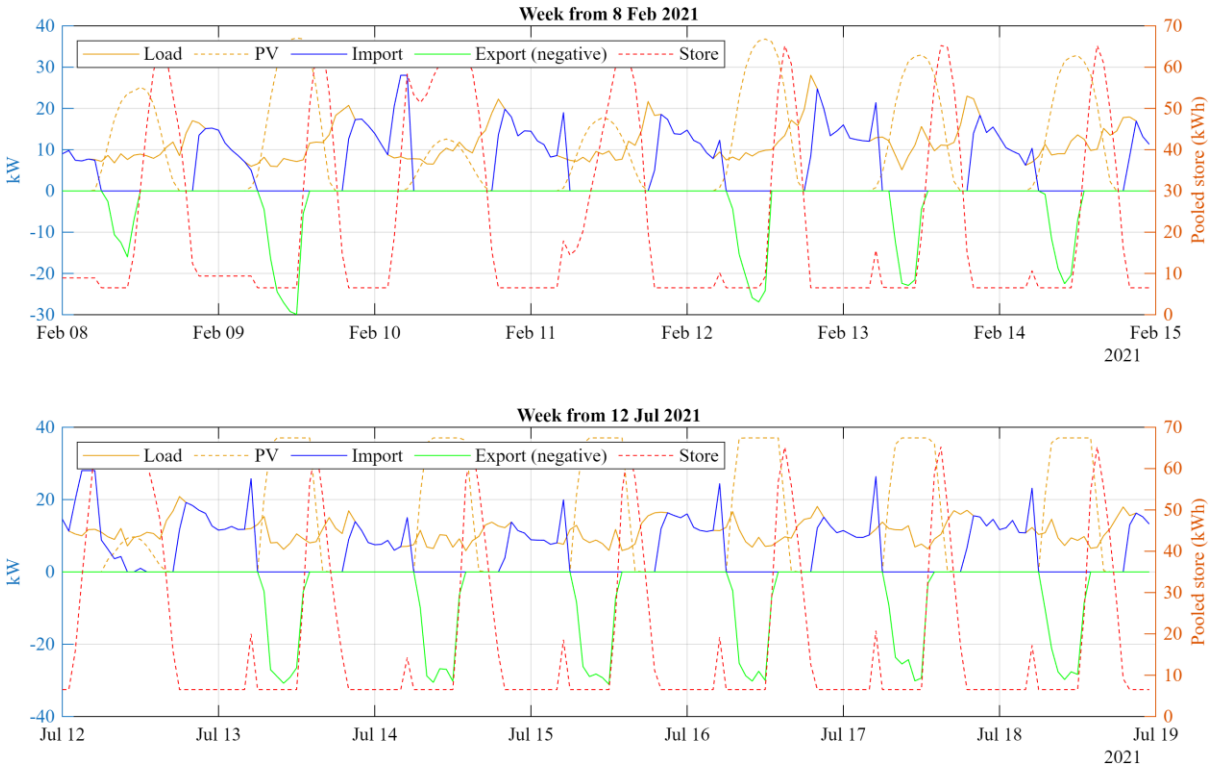


Figure 3. Hourly dispatch of the cooperative pool in a summer week (8 to 14 February) and a winter week (12 to 18 July) in the Base scenario: community load, solar generation, grid import, grid export (negative) and the state of charge of the shared store.

3.2 Households and their statements

Table **5** and Figure **4** give each household's bills and gain under the recommended rule. Every household gains. The smallest share is 16 % of the own bill (member 9: large load, small solar, no battery) and the smallest amount 157 AUD (member 6, already two-thirds self-sufficient alone). The two lower-income members, 2 and 5, gain 808 and 858 AUD, of which 233 AUD each is the solidarity share. Member 7 has no solar, so its 18.6 kWh battery is idle alone; in the pool it contributes the largest marginal saving and receives 1,569 AUD. Member 3, whose 13.2 kWp supplies the pool for most of the year, ends with a bill of 4 AUD. The nine statements in Appendix B (Table B**1** to Table B**10**) carry these numbers with each household's contribution and its Shapley share in words.

Table 5. Per-household bills and settlement under the recommended rule (AUD per year).

Member	Bill without assets	Stand-alone bill	Shapley share	Solidarity	Gain	Settled bill	Gain as share of stand-alone bill
1	1,720	685	222	0	200	485	29 %
2	1,509	1,509	638	233	808	701	54 %
3	2,881	289	316	0	285	4	99 %
4	2,212	686	294	0	265	421	39 %
5	2,237	2,237	694	233	858	1,379	38 %
6	2,613	563	175	0	157	405	28 %
7	3,399	3,399	1,744	0	1,569	1,830	46 %
8	2,944	1,039	218	0	196	843	19 %
9	3,097	2,018	365	0	328	1,690	16 %
All	**22,613**	**12,425**	**4,666**	**467**	**4,666**	**7,759**	**37.6 %**

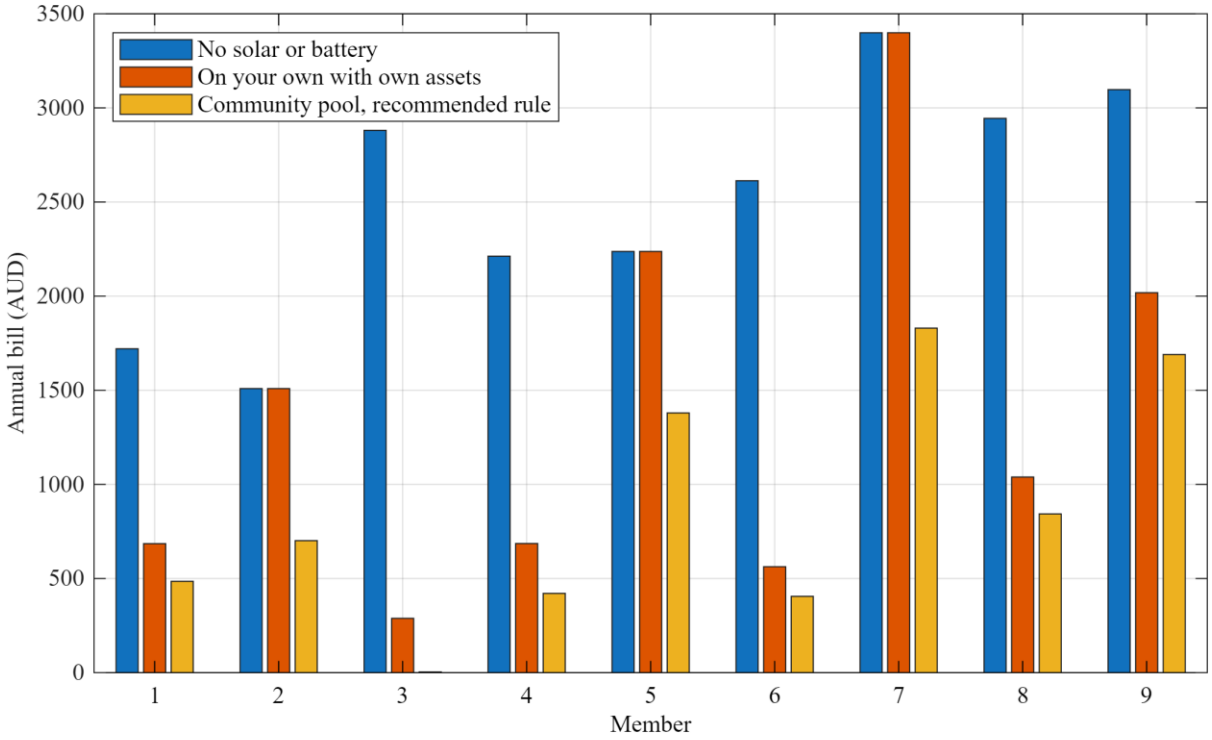


Figure 4. Annual bills of the nine households without assets, on their own with their own assets, and in the cooperative pool under the recommended rule (Base scenario).

3.3 Fairness of the allocation

Three rules are compared in Table **6**: the recommended rule, the Shapley value alone and the same-percentage baseline. Figure **5** shows the shares and Figure **6** the Lorenz curves. The recommended rule lifts the two lower-income members. It lowers the minimum share from 18 % (Shapley) to 16 %, because member 9 contributes little surplus and no storage. Its Gini coefficient of 0.290 is close to the Shapley value's 0.288 and far from the same-percentage rule's 0, which is zero by construction. The same-percentage rule would move 467 AUD from the battery and solar owners to the large consumers. It would give member 3 only 108 AUD for supplying 18.6 MWh of surplus, which is why it is a baseline and not the recommendation.

The stability checks show what the solidarity share costs. The Shapley allocation leaves one coalition able to gain 151 AUD by leaving the community; the recommended allocation raises that to 259 AUD, 5.6 % of the saving, because 467 AUD is moved away from the households that earned it at the margin. The nucleolus allocation (182, 655, 214, 259, 839, 89, 1,879, 126, 423 AUD for members 1 to 9) is the only one of the four inside the core,

with a maximum excess of −22 AUD. Stability has a price of its own: the nucleolus pays members 6 and 8 only 89 and 126 AUD, 12 % and 16 % of their bills, and gives member 7 the most. The recommended rule is retained because the 259 AUD margin is small against an annual saving of 4,666 AUD, every household still gains, and the two lower-income members are lifted by a rule that can be explained in one sentence.

Table 6. Fairness indicators of four allocation rules (Base scenario).

Rule	Minimum relative saving	Minimum gain (AUD)	Gini of relative savings	All households gain	Maximum core excess (AUD)
Recommended (Shapley with solidarity share)	0.163	157	0.290	yes	259
Shapley value alone	0.181	175	0.288	yes	151
Same percentage for everyone	0.376	108	0.000	yes	710
Nucleolus (stability check)	0.121	89	0.290	yes	-22

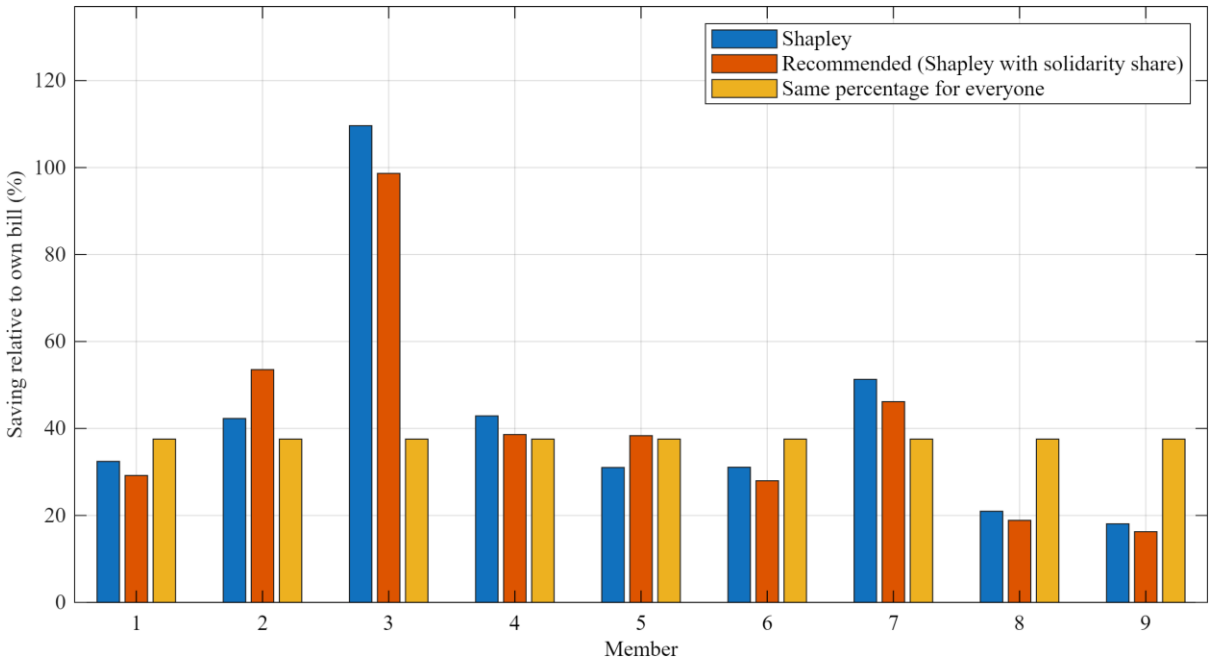


Figure 5. Saving relative to each household's own bill under the Shapley value, the recommended rule and the same-percentage baseline (Base scenario).

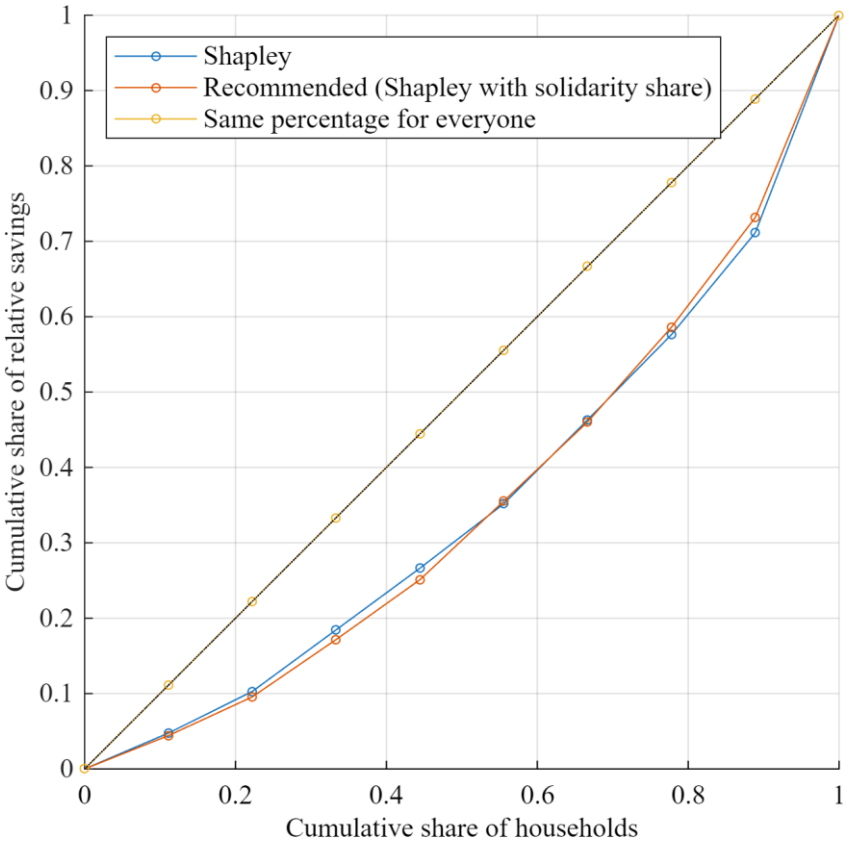


Figure 6. Lorenz curves of the relative savings under the three rules of Figure **5**; the diagonal is perfect equality of shares.

3.4 Grid and environmental performance

The natural cap holds the pooled import at the community's uncoordinated peak of 28.0 kW at no cost. Without it the optimiser would fill the store from the grid overnight at 0.15 AUD/kWh and raise the peak. The cap costs nothing because the cheapest hours are long enough to fill the store below 28 kW. Figure **7** shows the import duration curves and Figure **8** the price of a lower peak. An explicit cap of 25 kW costs 1 AUD per year, 20 kW costs 8 AUD and 15 kW costs 116 AUD, 1.5 % of the bill, with no unmet demand. In the safe mode the 15 kW cap is infeasible, because without grid charging the store cannot be pre-filled for winter evenings. The pool's import in the 0.55 window falls from 916 kWh under the simple rule to 63 kWh, a 93 % relief of the evening peak that the tariff is designed to flatten. The shared store completes 365 equivalent full cycles a year, one a day, within normal duty.

Self-consumption of 54 % and self-sufficiency of 60.5 % are the practical ceiling for this community. Its generation exceeds its load, so 38.4 MWh must be exported whatever the dispatch, and the remaining imports fall in winter evenings and nights when the store is empty. Emissions of imports are 22.6 t CO2-e per year at the 2021 NSW factor. A larger store would raise self-sufficiency, but the benchmark fixes the assets, so the environmental gain of the pool is the 38.4 MWh of clean export and the 853 kWh moved out of the peak window, not a lower total import.

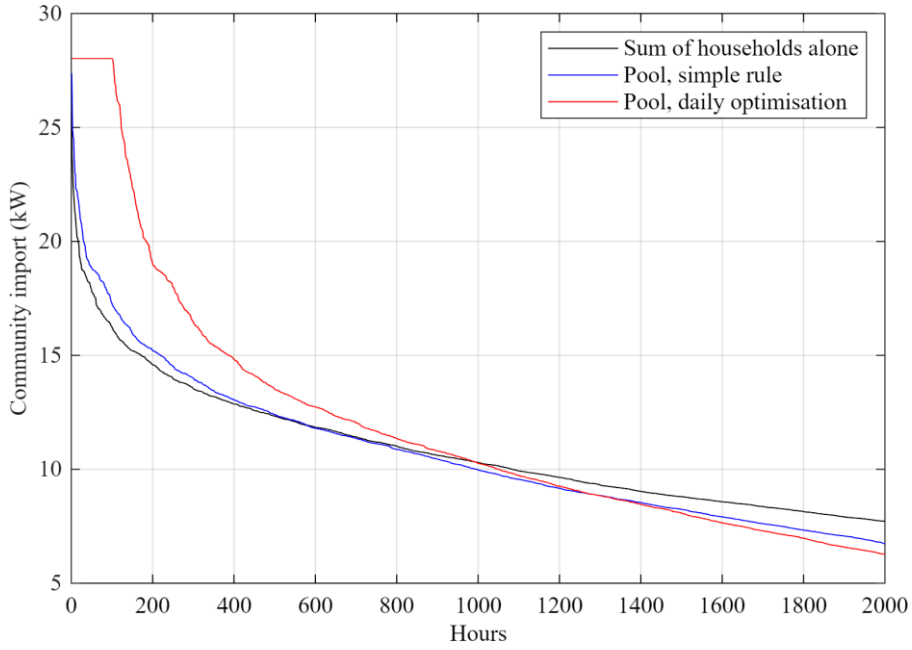


Figure 7. Duration curves of community import for the households alone, the pool under the simple rule and the pool under daily optimisation (first 2,000 hours).

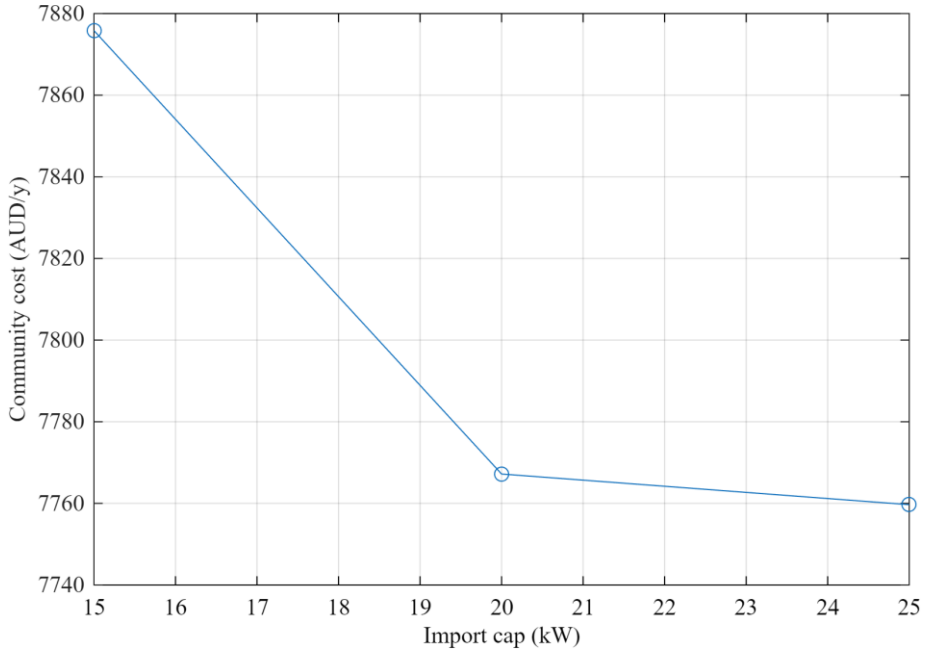


Figure 8. Community cost against an explicit import cap in the Base scenario; the natural cap of 28.0 kW costs nothing and a 15 kW cap costs 116 AUD per year.

3.5 Sensitivity

Figure **9** gives the one-at-a-time sensitivity with the self-consumption rule for all coalitions (base saving 4,238 AUD, Gini 0.297, minimum share 0.142). The import price level dominates: the saving runs from 2,351 to 6,124 AUD across 0.7 to 1.3 times the tariff. The cycling cost (4,238 to 2,542 AUD across 0.065 to 0.24 AUD/kWh) and the operator fee (4,688 to 3,068 AUD across 0 to 180 AUD per household per year) come next. The feed-in rate moves the Gini coefficient (0.222 to 0.366) more than the saving. The solar yield barely moves the saving, because generation already exceeds demand. At the upper ends of the cycling cost and the fee one household's gain falls to 3 % of its bill but stays positive. At 1.5 times the feed-in rate member 3's stand-alone bill turns negative and its relative saving is undefined, so absolute gains are the safer headline.

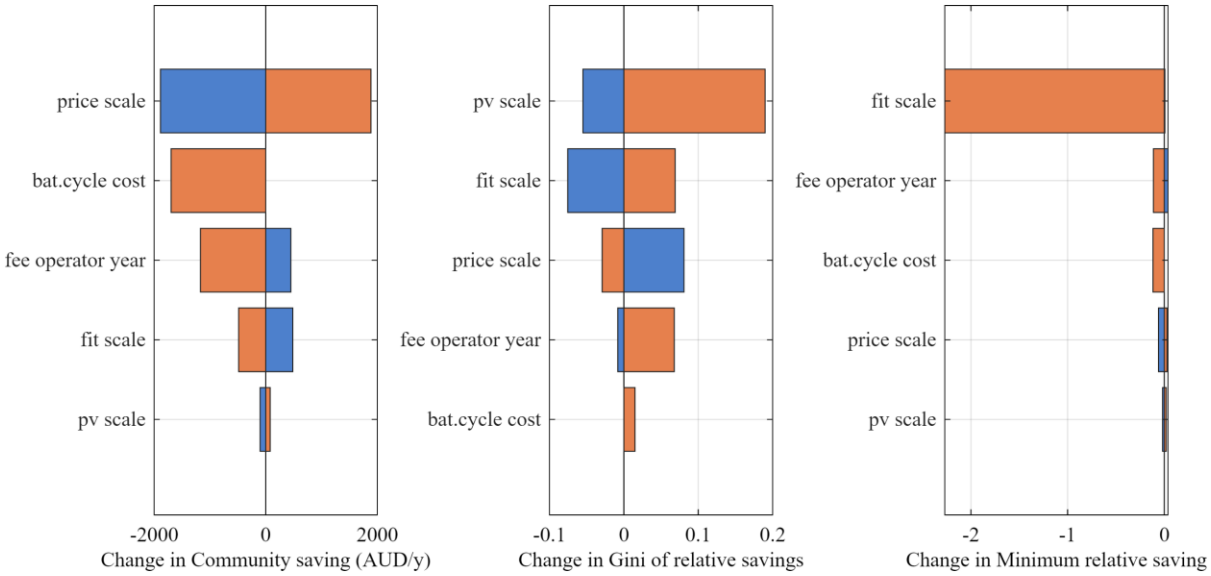


Figure 9. Tornado charts of the community saving, the Gini of relative savings and the minimum relative saving when each input moves to the low and high end of its range (self-consumption rule for all coalitions).

3.6 Practicality, durability and scalability

The pool needs what Australian community battery and virtual power plant trials already use: interval meters, one operator and a hybrid inverter per battery [23, 24, 25]. The operator fee of 50 AUD per household per year sits inside the range of those trials, from 0 to 180 AUD. Operation needs no forecast beyond the next day's load and solar. If grid charging is not permitted, the safe mode keeps 4,470 of the 4,666 AUD, 96 % of the saving. If no optimisation is available, the simple rule keeps 4,238 AUD, 91 %. Settlement is annual, which avoids the swings of monthly recomputation [4]. Durability rests on the cycling cost: every kWh discharged is charged 0.065 AUD for wear, so battery owners are paid for the use of their modules and the store completes one full cycle a day, within warranty duty [19]. Scaling changes one thing. The exact Shapley value needs every coalition, and the count doubles with each member: 511 coalitions and eleven minutes for nine households, about 64 times longer for fifteen. Above that size the published remedy is sampling of coalitions or clustering of similar members [5]. The dispatch, the import cap and the statements scale without change.

3.7 Transparency and reproducibility

All 46 inputs beyond the benchmark are listed with units, ranges, sources and notes in one workbook. The code reads every value from it at run time and saves a copy beside each run. The model is deterministic. It asserts the hourly energy balance and the state-of-charge window in every one of the 8,760 hours of every dispatch, and it reproduces all tables and figures with one command in about twelve minutes per scenario. The public repository contains the inputs workbook, the code, the headline results of the three scenarios as comma-separated files and the nine household statements; a full run regenerates the hourly dispatch, the 511 coalition values and the figures. Any number in this report can be traced to a file and recomputed (Appendix D).

4 Conclusion

Economic performance. The cooperative pool pays 7,759 AUD per year against 12,425 AUD for the nine households alone, a saving of 4,666 AUD or 37.6 %, and 65 % below the 22,613 AUD the households would pay without any assets. Of the saving, 428 AUD comes from the daily optimisation and 196 AUD from grid charging within the cap; the rest comes from pooling surplus solar with idle batteries. The saving is net of the operator's running cost.

Community benefit and energy justice. Every household gains, by 16 % to 99 % of its own bill and by 157 to 1,569 AUD. The guarantee that nobody pays more than alone is built into the definition of the gain. The two households with after-tax incomes below the Greater Sydney median receive 233 AUD each from the solidarity share on top of their marginal contributions. Each household can read on its statement how its share was computed: the average amount the saving grew when it joined, over every order in which the community could have formed.

Grid and environmental performance. The pool uses 54 % of its solar locally and meets 60.5 % of its demand locally, the ceiling for a community whose generation exceeds its load. Import in the 0.55 AUD/kWh window falls by 93 %, from 916 to 63 kWh. The pooled import never exceeds the community's own uncoordinated peak of 28.0 kW, at no cost, and a 15 kW cap is available for 116 AUD per year. Emissions of imports are 22.6 t CO2-e per year at the 2021 NSW factor.

Innovation. All 511 coalitions are dispatched with the same optimiser as the community, so the settlement rests on exact marginal contributions rather than on sampled or clustered approximations. The natural import cap turns a side effect of optimisation, overnight grid charging that raises the peak, into grid neutrality at zero cost. The whole model runs from the benchmark data and one inputs workbook.

Practicality, durability and scalability. The equipment is what Australian trials already install, and the operator fee sits inside their range. Without grid charging the pool keeps 96 % of its saving, and with the simple rule 91 %. Battery wear is paid at 0.065 AUD per kWh discharged, at one full cycle a day. The dispatch, the cap and the statements scale without change; the exact settlement is affordable to about fifteen members, beyond which sampling is needed.

Communication and reproducibility. Every household receives a one-page statement. Every number in this report traces to a comma-separated file in the public repository and is recomputed from the benchmark data with one command in about twelve minutes per scenario. Table C**1** maps each question of the challenge brief to the section that answers it.

Three limits remain. The benchmark gives no network data, so the community is a copper plate with no export limit. The calendar year is inferred from the tariff pattern. The exact Shapley value is limited to small communities.

Declarations

Data and code availability

The ComErgy Benchmark Dataset 2026 is provided by the organisers [1] and is obtained from the challenge page; it is not redistributed. The MATLAB code, the inputs workbook, the headline results of the three scenarios, the nine household statements and this report are published in a public repository under the MIT licence (code and workbook) and CC BY 4.0 (report): https://github.com/ED-Hammad/fairpool-sydney-comergy-2026 (release v1.0.0).

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Conflicts of interest

The author declares no conflict of interest.

Nomenclature

Symbol	Meaning	Unit
Sets and indices		
h	hour of the year, 1 to 8,760	
i	household index, 1 to 9	
N	the set of all nine households	
S	a coalition, any non-empty subset of N; in $S(h)$ and $S_i(h)$ the symbol denotes the state of charge	
$\|S\|$	number of households in coalition S	
W	planning window of the daily programme, 48 hours	
n_{low}	number of households whose income is below the income test	
Parameters		
$GHI, DNI,$	global horizontal, direct normal and diffuse horizontal irradiance	W/m2
POA	irradiance on the plane of the array	W/m2
θ	angle of incidence of beam irradiance on the panel	degrees
β	panel tilt, 30 degrees	degrees
ρ	ground albedo, 0.20	dimensionless
T	dry-bulb air temperature	C
$NOCT$	nominal operating cell temperature, 44 C	C
γ	power temperature coefficient, -0.0033	per K
η_{dc}, η_{inv}	DC loss factor, 0.86; inverter efficiency, 0.97	dimensionless
r	DC to AC ratio, 1.33	dimensionless
k_i	installed solar capacity of household i	kWp
L_i, L_S	load of household i; load of coalition S	kW
P_i, P_S	battery power limit of household i; of coalition S	kW
E_i^{min}, E_i^{max}, E_S^{min}, E_S^{max}	lower and upper state-of-charge bounds of household i and of coalition S	kWh
η	one-way battery efficiency, the square root of the round-trip efficiency 0.95	dimensionless
p, f	import price; feed-in rate	AUD/kWh
c_{cyc}	cycling cost, 0.065	AUD/kWh
c_{conn}	connection fee, 1	AUD/day
c_{op}	operator fee, 50	AUD/year
$\overline{I}_S$	natural import cap of coalition S, its own load peak	kW
σ	solidarity share, 0.10	dimensionless
y_i, y^*	after-tax income of household i; income test, 108,000	AUD
Variables		
y	solar generation per installed kilowatt	kWh per kWp per hour
T_c	cell temperature	C
G_i, G_S	solar generation of household i; of coalition S	kW
I, X	grid import; grid export (subscript i or S when specific)	kW
C, D	battery charge; battery discharge	kW
$S(h)$, $S_i(h)$	state of charge of the shared store; of the battery of household i	kWh
B_i^0, B_i	stand-alone bill; settled bill of household i	AUD/year
$c(S)$	pooled cost of coalition S including fees	AUD/year
$v(S), v(N)$	saving of coalition S; saving of the whole community	AUD/year
ϕ_i	Shapley value of household i	AUD/year
g_i	allocated gain of household i	AUD/year
$e^{max}(g)$	maximum core excess of allocation g	AUD/year

Symbol	Meaning	Unit
$\mathbb{1}\{\cdot\}$	indicator function, 1 when the condition holds and 0 otherwise	
Abbreviations		
AC, DC	alternating current, direct current	
ACT	Australian Capital Territory	
AUD	Australian dollar	
CEC	Clean Energy Council	
CO2-e	carbon dioxide equivalent	
IPART	Independent Pricing and Regulatory Tribunal of New South Wales	
kWp	kilowatt peak, installed solar capacity	
LFP	lithium iron phosphate (battery chemistry)	
NEM	National Electricity Market	
NMOT	nominal module operating temperature, the manufacturer's equivalent of NOCT	
NSW	New South Wales	
SDR	supply-demand ratio (illustrative hourly price)	

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Appendix A. Assumptions

Table A1. Assumptions beyond the benchmark data, with reasoning and sources.

ID	Assumption	Value	Reason	Sources
A1	Solar tilt and azimuth	30 degrees, due north	Published Sydney optimum 31 degrees; installer practice is the actual roof pitch and orientation, which the benchmark does not give	[7, 8, 9, 13]
A2	Albedo	0.20	Standard textbook value for ground without snow	[10]
A3	Module temperature coefficient and NOCT	-0.33 %/K; 44 C (manufacturer NMOT)	LG440N2W-E6 specification; the exact LG 440 W model in the benchmark is not named	[11]
A4	DC losses	14 % (factor 0.86)	PVWatts default loss stack; the CEC derating factors with the LG output warranty give 0.89 new and 0.86 at year 8 to 10	[11, 12, 13]
A5	Inverter efficiency and DC to AC ratio	0.97; 1.33	Fox ESS H3 Smart European efficiency 97.2 %; CEC limit that the inverter AC rating is at least 75 % of the array peak	[12, 13, 14]
A6	Irradiance model	Erbs decomposition, isotropic transposition, Spencer solar position	Standard, reproducible from the benchmark irradiance and temperature	[10, 31, 32]
A7	Battery power	0.48 kW per kWh (2.24 kW per module)	EQ4800 datasheet maximum 50 A per module; the paired inverter may cap lower	[6, 15]
A8	Round-trip efficiency	0.95 (battery pack)	Datasheet of the battery named in the benchmark; LFP packs 92 to 98 % in independent tests; inverter conversion losses are not charged to battery throughput	[6, 16]
A9	Usable window	10 to 100 %	Datasheet (September 2025): 100 % depth of discharge, usable equals nominal; a 10 % reserve is kept	[6, 16, 17]
A10	Initial state of charge	10 % on 1 January	Immaterial over a year	
A11	Cycling cost	0.065 AUD per kWh discharged	Lower end of the surveyed range (hardware after the federal rebate, excluding installation, over the warranted throughput of 4,219 kWh per kWh); the installed-cost value 0.18 cuts the community saving by 22 %	[16, 18, 19, 20]
A12	Operator fee	50 AUD per household per year	Inside the Australian trial range (0 at Ausgrid, 180 at the Origin Essential Energy trial); a fixed charge, a cost of every coalition	[22, 23, 24, 25, 26]
A13	Connection fee	1 AUD per household per day, retained	Benchmark value, read as per household	[1]
A14	Import price and feed-in rate	Benchmark values	The challenge fixes the tariff; IPART's 2021-22 benchmark corroborates the 5 c feed-in rate	[1, 21]
A15	Pooled dispatch	Linear programme solved each day over a two-day window, first day committed, full knowledge of the window	Standard community dispatch; the two-day window gives energy kept overnight a use	[2]
A16	Grid charging	Allowed in the pool, never in the stand-alone bill	The only way idle batteries of households without solar serve the pool	

ID	Assumption	Value	Reason	Sources
A17	Natural import cap	Pooled import never above the coalition's own uncoordinated load peak	Cooperation must not worsen the network peak	[33]
A18	Store aggregation	Five batteries as one store (65.2 kWh, 31.3 kW)	Exact for cost with identical modules	
A19	Stand-alone bill	Each household alone, own solar and battery, self-consumption rule, no grid charging	Disagreement point of cooperative allocation; what a hybrid inverter does out of the box	[3, 4, 5]
A20	Settlement rule	90 % by the exact Shapley value, 10 % solidarity split equally among members with after-tax income below 108,000 AUD (Greater Sydney median gross household income), floor at zero	Shapley is the axiomatic rule for small communities; the solidarity share is the justice component	[3, 4, 27, 34]
A21	Settlement cadence	Annual, one statement per household	Monthly recomputation is volatile	[4]
A22	Emission factor	0.79 kg CO2-e per kWh on imports	The challenge gives no factor; NSW and ACT scope 2 for 2021-22	[28, 29]
A23	Network	Copper plate within the community; no export limit	No network data in the benchmark	[1]

Appendix B. Household statements

Each household receives the statement below once a year. The numbers are those of the Base scenario (benchmark year 2021, AUD per year, energy in kWh). Table B**1** summarises the nine households; Table B**2** to Table B**10** give the individual statements. Energy "offered to the pool" is the household's solar beyond its own hourly use; "use not covered by own solar" is the energy the pool or the grid supplied. Import and export "alone" are what the household would exchange with the grid on its own with its own assets; in the pool the community exchanges energy with the grid through one meter, 28,665 kWh in and 38,435 kWh out. The contribution to peak reduction is given in two ways: the household's own import inside the 0.55 AUD/kWh window when alone, against a pool total of 63 kWh, and its load at the community's peak hour (12 February, 19:00, 28.0 kW). The pooled import never exceeds that peak.

Table B1. Summary of the nine household statements (Base scenario, AUD per year, energy in kWh).

Member	Stand-alone bill	Pooled share	Saving	Saving as share of own bill	Solar offered to the pool	Battery in the shared store	Import in the 0.55 window alone	Demand at the community peak (kW)
1	685	485	200	29 %	9,467	0	365	1.2
2	1,509	701	808	54 %	0	9.3	752	2.5
3	288	4	285	99 %	18,646	14.0	91	5.2
4	686	421	265	39 %	13,681	0	542	4.0
5	2,237	1,379	858	38 %	0	0	1,339	2.2
6	563	405	157	28 %	12,543	9.3	186	10.0
7	3,399	1,830	1,569	46 %	0	18.6	2,237	1.2
8	1,039	843	196	19 %	7,601	14.0	58	1.0
9	2,018	1,690	328	16 %	4,860	0	1,075	0.7
All	**12,425**	**7,759**	**4,666**	**37.6 %**	**66,798**	**65.2**	**6,644 (pool: 63)**	**28.0**

Member 1. Couple, no children, both adults working full-time (2 occupants). Annual statement for the benchmark year 2021 (Table B**2**).

Table B2. Statement of member 1 (Base scenario).

Item	Value
Electricity use	5,316 kWh
Solar generated by your panels	11,178 kWh (6.6 kWp, 15 modules)
Solar used at home in the same hour	1,711 kWh
Solar offered to the pool (beyond your own hourly use)	9,467 kWh
Use not covered by your own solar, supplied by the pool or the grid	3,605 kWh
Battery placed in the shared store	none (no battery installed)
Grid import and export if you were alone	3,605 and 9,467 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	365 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	4.3 kW; 1.2 kW
Bill with no solar or battery (for context)	1,720 AUD
Bill on your own with your own assets (your guaranteed ceiling)	685 AUD
Your share of the pooled community bill	485 AUD
Your saving from the pool	200 AUD (29 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	222 AUD (4.8 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	257 AUD

Your pooled share of 485 AUD is below your own bill of 685 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 365 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 2. Retired couple (2 occupants). Annual statement for the benchmark year 2021 (Table B**3**).

Table B3. Statement of member 2 (Base scenario).

Item	Value
Electricity use	4,401 kWh
Solar generated by your panels	none (no solar installed)
Solar used at home in the same hour	0 kWh
Solar offered to the pool (beyond your own hourly use)	0 kWh
Use not covered by your own solar, supplied by the pool or the grid	4,401 kWh
Battery placed in the shared store	9.3 kWh (2 modules)
Energy your battery would charge and discharge alone	0 and 0 kWh (0 full cycles)
Grid import and export if you were alone	4,401 and 0 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	752 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	4.0 kW; 2.5 kW
Bill with no solar or battery (for context)	1,509 AUD
Bill on your own with your own assets (your guaranteed ceiling)	1,509 AUD
Your share of the pooled community bill	701 AUD
Your saving from the pool	808 AUD (54 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	638 AUD (13.7 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	233 AUD
What the same-percentage rule would give you, for comparison	567 AUD

Your pooled share of 701 AUD is below your own bill of 1,509 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share plus 233 AUD from the solidarity share. On your own you would import 752 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 3. Couple with children, one adult employed full-time and one stay-at-home parent (5 occupants). Annual statement for the benchmark year 2021 (Table B**4**).

Table B4. Statement of member 3 (Base scenario).

Item	Value
Electricity use	9,260 kWh
Solar generated by your panels	22,356 kWh (13.2 kWp, 30 modules)
Solar used at home in the same hour	3,711 kWh
Solar offered to the pool (beyond your own hourly use)	18,646 kWh
Use not covered by your own solar, supplied by the pool or the grid	5,549 kWh
Battery placed in the shared store	14.0 kWh (3 modules)
Energy your battery would charge and discharge alone	2,493 and 2,358 kWh (187 full cycles)
Grid import and export if you were alone	3,192 and 16,152 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	91 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	7.4 kW; 5.2 kW
Bill with no solar or battery (for context)	2,881 AUD
Bill on your own with your own assets (your guaranteed ceiling)	288 AUD
Your share of the pooled community bill	4 AUD
Your saving from the pool	285 AUD (99 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	316 AUD (6.8 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	108 AUD

Your pooled share of 4 AUD is below your own bill of 288 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 91 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 4. Couple with children, both adults employed full-time (4 occupants). Annual statement for the benchmark year 2021 (Table B**5**).

Table B5. Statement of member 4 (Base scenario).

Item	Value
Electricity use	7,131 kWh
Solar generated by your panels	16,395 kWh (9.7 kWp, 22 modules)
Solar used at home in the same hour	2,714 kWh
Solar offered to the pool (beyond your own hourly use)	13,681 kWh
Use not covered by your own solar, supplied by the pool or the grid	4,417 kWh
Battery placed in the shared store	none (no battery installed)
Grid import and export if you were alone	4,417 and 13,681 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	542 kWh

Item	Value
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	6.9 kW; 4.0 kW
Bill with no solar or battery (for context)	2,212 AUD
Bill on your own with your own assets (your guaranteed ceiling)	686 AUD
Your share of the pooled community bill	421 AUD
Your saving from the pool	265 AUD (39 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	294 AUD (6.3 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	258 AUD

Your pooled share of 421 AUD is below your own bill of 686 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 542 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 5. Single-parent family with children, one adult part-time employed (3 occupants). Annual statement for the benchmark year 2021 (Table B**6**).

Table B6. Statement of member 5 (Base scenario).

Item	Value
Electricity use	6,921 kWh
Solar generated by your panels	none (no solar installed)
Solar used at home in the same hour	0 kWh
Solar offered to the pool (beyond your own hourly use)	0 kWh
Use not covered by your own solar, supplied by the pool or the grid	6,921 kWh
Battery placed in the shared store	none (no battery installed)
Grid import and export if you were alone	6,921 and 0 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	1,339 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	3.5 kW; 2.2 kW
Bill with no solar or battery (for context)	2,237 AUD
Bill on your own with your own assets (your guaranteed ceiling)	2,237 AUD
Your share of the pooled community bill	1,379 AUD
Your saving from the pool	858 AUD (38 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	694 AUD (14.9 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	233 AUD
What the same-percentage rule would give you, for comparison	840 AUD

Your pooled share of 1,379 AUD is below your own bill of 2,237 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share plus 233 AUD from the solidarity share. On your own you would import 1,339 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 6. Couple with children, one adult full-time and one part-time employed (5 occupants). Annual statement for the benchmark year 2021 (Table B**7**).

Table B7. Statement of member 6 (Base scenario).

Item	Value
Electricity use	7,436 kWh
Solar generated by your panels	14,904 kWh (8.8 kWp, 20 modules)
Solar used at home in the same hour	2,361 kWh
Solar offered to the pool (beyond your own hourly use)	12,543 kWh
Use not covered by your own solar, supplied by the pool or the grid	5,075 kWh
Battery placed in the shared store	9.3 kWh (2 modules)
Energy your battery would charge and discharge alone	2,930 and 2,784 kWh (332 full cycles)
Grid import and export if you were alone	2,292 and 9,613 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	186 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	10.0 kW; 10.0 kW
Bill with no solar or battery (for context)	2,613 AUD
Bill on your own with your own assets (your guaranteed ceiling)	563 AUD
Your share of the pooled community bill	405 AUD
Your saving from the pool	157 AUD (28 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	175 AUD (3.7 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	211 AUD

Your pooled share of 405 AUD is below your own bill of 563 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 186 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 7. Multi-generation home, two adults retired and two employed full-time (7 occupants). Annual statement for the benchmark year 2021 (Table B**8**).

Table B8. Statement of member 7 (Base scenario).

Item	Value
Electricity use	10,748 kWh
Solar generated by your panels	none (no solar installed)
Solar used at home in the same hour	0 kWh
Solar offered to the pool (beyond your own hourly use)	0 kWh
Use not covered by your own solar, supplied by the pool or the grid	10,748 kWh
Battery placed in the shared store	18.6 kWh (4 modules)
Energy your battery would charge and discharge alone	0 and 0 kWh (0 full cycles)
Grid import and export if you were alone	10,748 and 0 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	2,237 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	5.7 kW; 1.2 kW
Bill with no solar or battery (for context)	3,399 AUD
Bill on your own with your own assets (your guaranteed ceiling)	3,399 AUD
Your share of the pooled community bill	1,830 AUD
Your saving from the pool	1,569 AUD (46 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	1,744 AUD (37.4 % of the community saving)

Item	Value
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	1,276 AUD

Your pooled share of 1,830 AUD is below your own bill of 3,399 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 2,237 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 8. Couple with children, one adult full-time and one part-time employed (6 occupants). Annual statement for the benchmark year 2021 (Table B**9**).

Table B9. Statement of member 8 (Base scenario).

Item	Value
Electricity use	10,463 kWh
Solar generated by your panels	11,178 kWh (6.6 kWp, 15 modules)
Solar used at home in the same hour	3,577 kWh
Solar offered to the pool (beyond your own hourly use)	7,601 kWh
Use not covered by your own solar, supplied by the pool or the grid	6,886 kWh
Battery placed in the shared store	14.0 kWh (3 modules)
Energy your battery would charge and discharge alone	3,529 and 3,353 kWh (266 full cycles)
Grid import and export if you were alone	3,533 and 4,072 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	58 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	6.9 kW; 1.0 kW
Bill with no solar or battery (for context)	2,944 AUD
Bill on your own with your own assets (your guaranteed ceiling)	1,039 AUD
Your share of the pooled community bill	843 AUD
Your saving from the pool	196 AUD (19 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	218 AUD (4.7 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	390 AUD

Your pooled share of 843 AUD is below your own bill of 1,039 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 58 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Member 9. Share house, all adults full-time students with part-time or casual work (7 occupants). Annual statement for the benchmark year 2021 (Table B**10**).

Table B10. Statement of member 9 (Base scenario).

Item	Value
Electricity use	10,891 kWh
Solar generated by your panels	7,452 kWh (4.4 kWp, 10 modules)
Solar used at home in the same hour	2,592 kWh
Solar offered to the pool (beyond your own hourly use)	4,860 kWh

Item	Value
Use not covered by your own solar, supplied by the pool or the grid	8,299 kWh
Battery placed in the shared store	none (no battery installed)
Grid import and export if you were alone	8,299 and 4,860 kWh
Your import in the 0.55 AUD/kWh window if you were alone (the pool imports 63 kWh in total)	1,075 kWh
Your highest hourly demand; your demand at the community peak hour (28.0 kW)	7.1 kW; 0.7 kW
Bill with no solar or battery (for context)	3,097 AUD
Bill on your own with your own assets (your guaranteed ceiling)	2,018 AUD
Your share of the pooled community bill	1,690 AUD
Your saving from the pool	328 AUD (16 % of your own bill)
Your Shapley share (average amount the saving grew when you joined, over all 511 groups)	365 AUD (7.8 % of the community saving)
Solidarity share (10 % of the community saving, split between the lower-income members)	not applicable
What the same-percentage rule would give you, for comparison	758 AUD

Your pooled share of 1,690 AUD is below your own bill of 2,018 AUD, so you are not worse off; this holds for every household by construction. Your gain is 90 % of your Shapley share; the other 10 % of the community saving forms the solidarity share for the two lower-income members. On your own you would import 1,075 kWh in the 0.55 AUD/kWh window; the pool imports 63 kWh in that window for all nine households. Its highest draw from the grid, 28.0 kW, is no higher than the community would have had without the pool.

Appendix C. Where the challenge questions are answered

Table C1. Questions of the challenge brief [1], the section that answers each, and the answer in one line.

Challenge question	Section	Answer in one line
How electricity is generated, stored and exchanged within the community	2.2, 2.4	Rooftop solar from the benchmark weather serves all loads first; the fourteen battery modules act as one shared store; surplus and deficit are pooled through a single virtual meter
How electricity is purchased from and exported to the grid	2.4, 3.1	A daily linear programme imports at the time-of-use price and exports at the feed-in rate, with import never above the community's own peak; 28,665 kWh in, 38,435 kWh out
How local energy trading operates	2.5, 2.7	No hourly bilateral trades; the annual saving of 4,666 AUD is settled once a year by the Shapley value with a solidarity share
How local electricity prices are determined	2.7	An illustrative hourly supply-demand-ratio price between the feed-in rate and the import price is published; settlement does not depend on it
How rooftop solar and batteries are coordinated	2.4, 3.1	One optimiser schedules the shared store two days ahead and commits one day; worth 428 AUD per year over the simple rule
How household and community batteries are coordinated	2.4, 3.6	The five household batteries form the community store; there is no separate community battery, and idle batteries of households without solar serve the pool
How the financial and broader benefits are allocated	2.5, 3.2, 3.3	Exact Shapley value over 511 coalitions, 10 % solidarity share to members 2 and 5, floor at zero; stability checked against the core and the nucleolus
How no household is disadvantaged	2.3, 3.2	Each gain is measured against the stand-alone bill, so the settled bill is never higher; the smallest gain is 16 % of the own bill

Challenge question	Section	Answer in one line
How the local market supports the wider grid	2.4, 3.4	The import cap keeps the peak at 28.0 kW at no cost, a 15 kW cap costs 116 AUD per year, and peak-window import falls by 93 %
How the mechanism stays transparent, trustworthy and practical	2.7, 3.6, 3.7, Appendix B	One-page statements, an open inputs workbook, deterministic code with hourly assertions, equipment and fees from Australian trials

Appendix D. Reproduction

The repository contains the inputs workbook, the code and the headline results; the benchmark workbook is downloaded from the challenge page into the data folder. To reproduce the Base scenario in MATLAB R2025b with the Optimization Toolbox, open the repository folder, add the code folder to the path and run the driver with the scenario name:

```
addpath(fullfile(pwd,'code')); gc_main('full','Base')
```

The other scenarios are 'NoGridCharging' and 'RuleDispatch'. A full run takes about twelve minutes. The quick mode, gc_main('quick','Base'), runs in one minute with the simple rule for every coalition. Each run writes its results and figures with a run tag and saves a copy of the inputs it used.